Math 421

Wednesday, December 10

Announcements

- This week's drop-in hours –
  - **Today** 12:00-1:00 pm VV B205
  - Thursday 9:00 – 11:00 am VV B205
- Final Exam –
  - Date: Monday, December 15^{th}
  - Time: 5:05-7:05 pm
  - Location: Bardeen 140
- Course Evaluations!
  - Open until 11:59pm **today!**

Continuity ⇒ Integrability

Theorem 33.2. Every continuous function f on $[a, b]$ is integrable.

Proof. Let $\epsilon > 0$. Since f is continuous on $[a, b]$, it is uniformly continuous on every closed subinterval, so there exists $\delta > 0$ such that for all $x, y \in [a, b]$ if $|x - y| < \delta$ then $|f(x) - f(y)| < \boxed{\frac{\varepsilon}{b-a}}$.

Let P be a partition of $[a, b]$ such that $t_k - t_{k-1} < \delta$. By the Extreme Value Theorem, on each $[t_k, t_{k-1}]$ there exists x_k and y_k such that $M_k = f(x_k) = \sup\{f(x) : x \in [t_{k-1}, t_k]\}$ and $m_k = f(y_k) = \inf\{f(x) : x \in [t_{k-1}, t_k]\}$. Then

$$U(f,P) - L(f,P) = \sum_{k=1}^{n} (M_k - m_k)(t_k - t_{k-1})$$

$$= \sum_{k=1}^{n} \big(f(x_k) - f(y_k)\big)(t_k - t_{k-1})$$

$$< \boxed{\frac{\varepsilon}{b-a}} \sum_{k-1}^{n} (t_k - t_{k-1})$$

$$< \boxed{\frac{\varepsilon}{b-a}} (b - a) = \epsilon$$

Fundamental Theorem of Calculus I

$\to$ MVT

$\to$ IVT

Theorem 24.1. If g is a continuous function on $[a, b]$ that is differentiable on (a, b), and if g' is integrable on $[a, b]$, then

$$\boxed{\int_a^b g'} = g(b) - g(a)$$

$\to$ cauchy criterion for integrability (ε, P)

$L(g', P) \le \int_a^b g' \le U(g', P)$

$\to \left|\int_a^b g' - [g(b) - g(a)]\right| < \varepsilon$

Proof. Let $\varepsilon > 0$. Since g' is integrable on $[a,b]$, by the Cauchy criterion for integrability, there exists $P = \{a, t_1, \dots, t_{n-1}, b\}$ such that $U(g', P) - L(g', P) < \varepsilon$.

Since g is continuous on $[a,b]$ and differentiable on (a,b), by the MVT, there exists $x_k \in (t_{k-1}, t_k) \subseteq (a,b)$ such that

$$g'(x_k) = \frac{g(t_k) - g(t_{k-1})}{t_k - t_{k-1}} \implies \overbrace{\underbrace{g'(x_k)}_{\ge\, m_k}}^{\le\, M_k} (t_k - t_{k-1}) = g(t_k) - g(t_{k-1})$$

$$g(b) - g(a) = \sum_{k=1}^{n} \left[g(t_k) - g(t_{k-1}) \right] = \sum_{k=1}^{n} g'(x_k)(t_k - t_{k-1})$$

$$L(g', P) \le g(b) - g(a) \le U(g', P)$$

Proof (cont.)

① $L(g', P) \leq \int_a^b g' \leq U(g', P)$

② $L(g', P) \leq g(b) - g(a) \leq U(g', P)$

$-U(g', P) \leq -[g(b) - g(a)] \leq -L(g', P)$

① - ② : $L(g', P) - U(g', P) \leq \left(\int_a^b g'\right) - [g(b) - g(a)] \leq U(g', P) - L(g', P)$

$\Rightarrow \left| \int_a^b g' - [g(b) - g(a)] \right| \leq U(g', P) - L(g', P) < \varepsilon$

Since $\varepsilon > 0$ was arbitrary $\int_a^b g' = g(b) - g(a)$.

* Properties of the Integral from § 33.

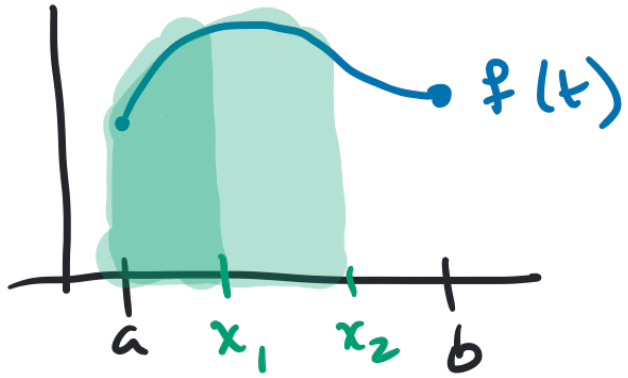
f is bounded

Fundamental Theorem of Calculus II

Theorem 34.3. Let f be an integrable function on $[a, b]$. For $x \in [a, b]$, let

Accumulation Function:

$$F(x) = \int_a^x f(t)\,dt$$



① → prove unif. cont. using ε, δ def.

② ↳ ε, δ def of cont.

Then F is continuous on $[a, b]$. If f is continuous at x_0 in (a, b), then F is differentiable at x_0 and

$$F'(x_0) = f(x_0)$$

$\lim_{x \to x_0} \dfrac{F(x) - F(x_0)}{x - x_0} = f(x_0)$

$\forall\, \varepsilon > 0,\ \exists\, \delta > 0$ s.t. $x, x_0 \in [a,b]$, $|x - x_0| < \delta$

$\Rightarrow \left| \dfrac{F(x) - F(x_0)}{x - x_0} - f(x_0) \right| < \varepsilon.$

Proof. Let $\varepsilon > 0$. Since f is integrable on $[a,b]$, f is bounded on $[a,b]$ so there exists $M > 0$ such that $|f(x)| \le M$, for all $x \in [a,b]$.

Assume $x \le y \in [a,b]$, and $|x-y| < \boxed{\frac{\varepsilon}{M}}$, then

$$|F(y) - F(x)| = \left| \int_a^y f(t)\,dt - \int_a^x f(t)\,dt \right|$$

$$= \left| \int_x^y f(t)\,dt \right| \le \int_x^y |f(t)|\,dt$$

$$\le \int_x^y M\,dt = M(y-x) = M|x-y| < \varepsilon$$

Since $\varepsilon > 0$ was arbitrary, F is uniformly continuous, and therefore continuous on $[a,b]$.